

by unpredictable actors once unleashed online, regardless of intent. Both BlackEnergy and WannaCry represent how malware proliferation risks spiralling out of control due to availability of state-grade arsenals and difficulty of attribution. Just as unrestrained proliferation of physical weapons risks fuelling conflict, leakage of cyber tools into unaccountable hands precipitates chaos. Tighter oversight and use policies around cyber weapons are necessary to prevent dangerous malware from going rogue.

Effective cyber resilience and precautionary measures

Proactive monitoring, system redundancies, effective contingency planning, and truly comprehending cyber risks are crucial to limit the impacts of events like Stuxnet, BlackEnergy, and WannaCry when they inevitably arise. Preparedness combined with international cooperation and clear cyber warfare deterrence policies can enhance stability even between state adversaries.

On the technical level, crucial precautions and controls include:

- Prioritising patching, upgrades and configuration management to eliminate vulnerabilities. EternalBlue and other leaked NSA Windows exploits utilised by WannaCry were patched by Microsoft but not deployed.
- Segmenting and firewalling networks and air-gapping sensitive systems to contain threats and prevent lateral movement. Stuxnet highlighted risks of any connectivity.
- Diversifying technology providers in the supply chain while vetting third parties extensively. Single points of failure invite compromise.
- Implementing multi-factor authentication and principle of least privilege to prevent post-breach lateral movement.
- Developing layered defences

combining tools like intrusion detection systems and antivirus software with threat intelligence.

- Regularly backing up critical data offline to limit ransomware impact and aid recovery.
- Establishing detailed cyber incident response plans through exercises. Contingencies are key to mitigating events.
- Continuously monitoring networks for anomalies and early threat detection.

From a human perspective, meticulous access controls, insider threat monitoring, and extensive staff security training are vital. Experts estimate 95% of breaches involve some human error. Common oversights include weak passwords, clicking phishing links, misconfigured cloud services, and mishandling sensitive data.

Preventative training, role-based access policies, and robust monitoring of personnel activity help address these errors. With strong preparedness and contingency planning, even sophisticated attacks can be isolated, contained, and recovered from. But implementation requires diligent leadership and cybersecurity maturity over time.

On a policy level, several best practices also emerge:

- Responsible disclosure of confirmed state-sponsored threats by governments and companies is important for collective awareness and action.
- Cyber warfare deterrence doctrines must be clearly communicated to avoid miscalculation and uncontrolled escalation.
- International norms around proportionality, critical infrastructure protection, and civilian impact need development through multilateral treaties and agreements.
- Information sharing and contingency planning between public and private sectors are essential to

manage cyber events.

- Resilient backup systems, emergency protocols, and staff training for critical infrastructure allow maintaining essential functions despite disruptions.

- With collaboration and open but secure communications around threats, even state adversaries can enhance stability and avoid worst-case cyber warfare scenarios.

Illustrative case studies

Major cyber incidents around the world highlight the high cost of inadequate security and preparedness as well as the path toward resilience:

Maersk. The shipping giant suffered \$300 million in damages after the NotPetya malware encrypted data and destroyed operations in 2017 due to unpatched systems and lack of backups. This highlighted global supply chain risks.

Target. The company lost data for 110 million customers in 2013 due to poor network segmentation that allowed lateral movement alongside ignoring threat alerts.

USA OPM. 21 million government employee records were compromised by Chinese state hackers in 2015 after failing to patch vulnerabilities and inadequate internal security practices that enabled long-term access.

Equifax. 143 million consumer records containing highly sensitive information were stolen in 2017 following unpatched systems and lack of segmentation that allowed uncontrolled internal access.

SolarWinds. In 2020 hackers inserted malware into trusted software updates to breach government agencies and companies worldwide. Showed risks of supply chain compromise.

In India, major incidents included:

Aadhaar. Reports emerged in 2017 of over 200 government websites allegedly leaking Aadhaar